Fin 203 - Final - ANOVA

Thursday, December 7, 2017 11:38 AM

Prob 1: $n = 30$

$$F = 4.9 = \frac{MSR}{MSE} = \frac{SSR/dfR}{SSE/df_e} = \frac{SSR}{SSE} \frac{24}{5}$$

$$\Downarrow$$

$$\frac{SSE}{SSR} = \frac{24}{4.9 \times 5}$$

$$MSR = \frac{SSR}{dfR} \qquad dfR = k = 5$$

$$MSE = \frac{SSE}{df_e} \qquad df_e = n - (k+1) = 24$$

$$R^2 = \frac{SSR}{SST} = \frac{1}{1 + \frac{SSE}{SSR}} = \frac{1}{1 + \frac{24}{4.9 \times 5}} \checkmark$$

$$\bar{R}^2 = 1 - \frac{SSE}{SST} \times \frac{n-1}{n-(k+1)} = 1 - \frac{1}{1 + \left(\frac{SSE}{SSR}\right)^{-1}} \times \frac{n-1}{n-(k+1)}$$

$$= 1 - \frac{1}{1 + \left(\frac{24}{4.9 \times 5}\right)^{-1}} \times \frac{29}{24} \checkmark$$